



UNIVERSITÀ
DEGLI STUDI
DI PADOVA



Dipartimento
di Fisica
e Astronomia
Galileo Galilei



UNIVERSITÀ DI PADOVA
Dipartimento
di Matematica
"Tullio Levi-Civita"

Network analysis of earthquakes in Italy

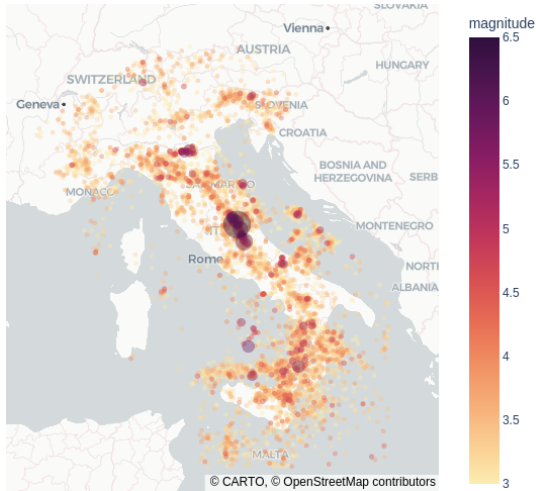
Benedetto Secco
Andrea Valsecchi

2nd July 2026

Dataset

1. Earthquake data were obtained from the **INGV catalog** (Istituto Nazionale di Geofisica e Vulcanologia), focusing on the Italian region over the period 1985–2025, for a total of 215,679 events.
2. A **spatial filtering** was applied, retaining only events within a customized polygonal area covering the Italian peninsula, the Alps, and nearby offshore regions (up to ~ 100 km from the coastline), resulting in 209,680 events.
3. The main variables considered are:
 - time
 - latitude
 - longitude
 - depth
 - magnitude

Earthquakes distribution



Here on the side, we see the scatter plot of the earthquakes (filtered for $\text{Mag} > 3$). The size and colour are scaled following the magnitude.

Gutenberg–Richter Law

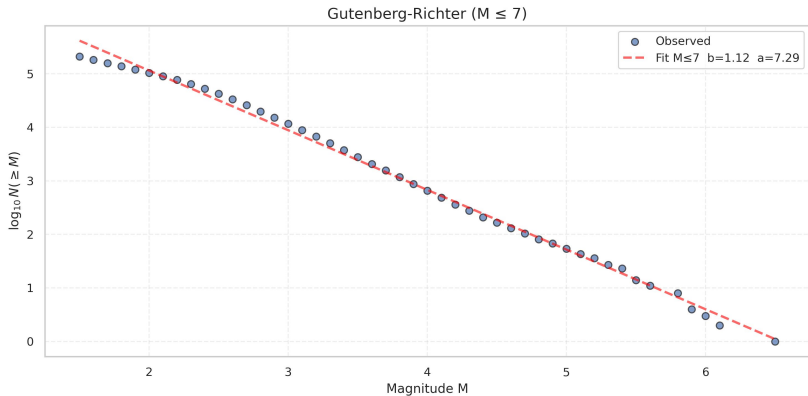
The Gutenberg–Richter law describes the frequency distribution of earthquake magnitudes.

- Empirical observation: small earthquakes are much more frequent than large ones
- Mathematical form:

$$\log_{10} N(M) = a - bM$$

- where:
 - $N(M)$: number of earthquakes with magnitude $\geq M$
 - a : overall seismic activity level (higher $a \Rightarrow$ more active region)
 - b : relative proportion of small vs large events
 - typically $b \approx 1$, which means that for each unit increase in magnitude, the number of earthquakes decreases by a factor of about 10

GR Law fit



Omori's Law (Aftershock Decay)

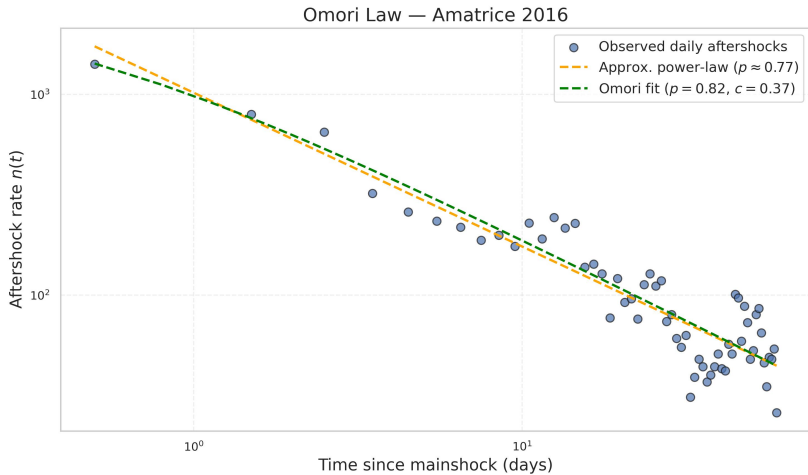
Omori's law (1894) describes how the rate of aftershocks decreases after a main shock. The generalized form (Utsu, 1961) is given by:

$$n(t) = \frac{K}{(c + t)^p}$$

where:

- $n(t)$: aftershock rate (number per unit time) at time t
- K, c : empirical constants that vary between different earthquake sequences
- p : decay exponent ($0.7 \lesssim p \lesssim 1.5$)
 - $p > 1$: fast decay (warm / fluid-rich crust)
 - $p < 1$: slow decay (cold / dry crust)

Omori Law fit



Abe–Suzuki Network Model

The Abe–Suzuki model (early 2000s) is one of the first approaches to represent seismic activity as a network.

- The earthquake catalog is mapped into a **directed network**:
 - Each **node** represents a spatial 3D cell (region discretization).
 - Each **edge** encodes temporal succession between events.
- For two consecutive earthquakes:
 - Let event i occur in cell A and event $i + 1$ in cell B .
 - A directed edge $A \rightarrow B$ is created.
- **Edge weights** represent the number of transitions between pairs of cells.
- The resulting network captures **spatio-temporal correlations** in seismic activity.

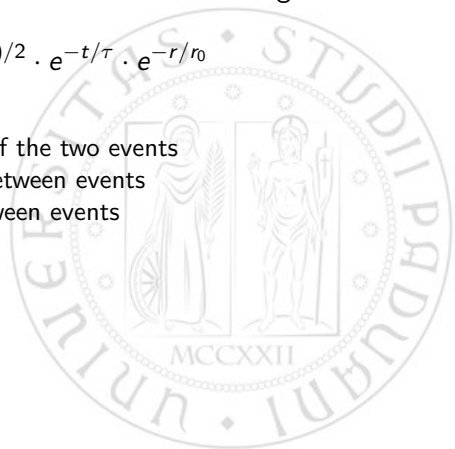
Our Custom Network: Weights

We extend the Abe–Suzuki model by introducing a **physically motivated weighting scheme**.

- Each directed edge between consecutive events is assigned a weight:

$$w_{ij} \propto 10^{\alpha(M_i+M_j)/2} \cdot e^{-t/\tau} \cdot e^{-r/r_0}$$

- where:
 - M_i, M_j are the magnitudes of the two events
 - t is the temporal distance between events
 - r is the spatial distance between events
- Model parameters:
 - $\alpha = 0.7$ (magnitude weight)
 - $\tau = 12$ h (temporal decay)
 - $r_0 = 30$ km (spatial decay)

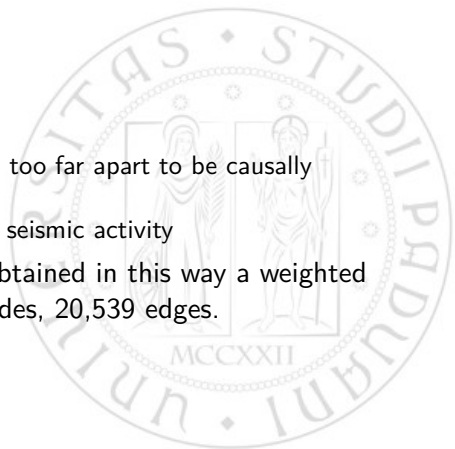


Our Custom Network: Cutoffs

To ensure physical relevance, we introduce also **hard cutoffs** in time and space.

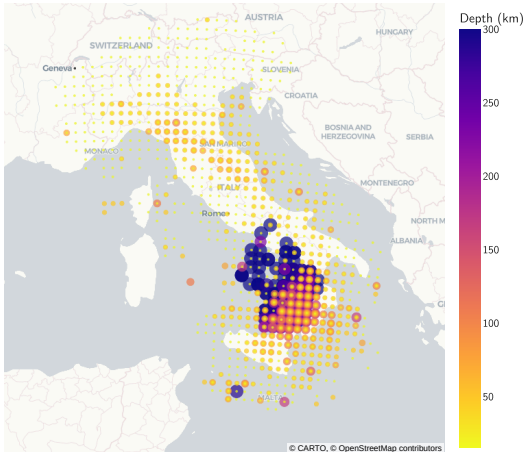
- Edges are created only if:
 - $t < t_{\max}$, $t_{\max} = 3 \text{ days}$
 - $r < r_{\max}$, $r_{\max} = 300 \text{ km}$
- Motivation:
 - Avoid linking events that are too far apart to be causally related
 - Reduce noise from unrelated seismic activity

Using a cell size of $a = 30 \text{ km}$, we obtained in this way a weighted and directed network with 1,798 nodes, 20,539 edges.



Earthquakes Depth

Cell depth



Cell depth.

The deep (~ 300 km) cluster in southern Italy is the Calabrian Arc subduction slab.

Hubs with DISS faults

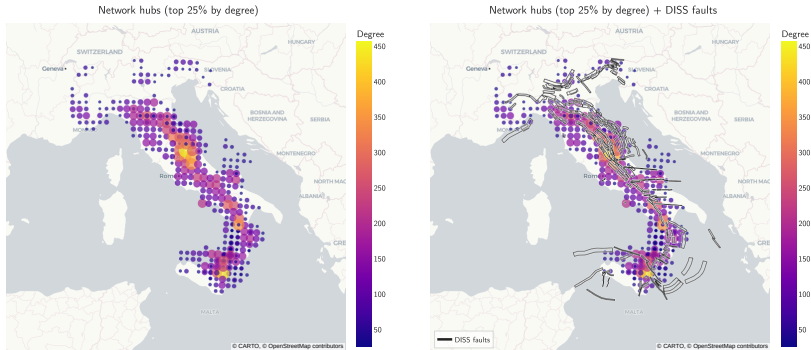


Figure: Top hubs (by degree) with and without DISS faults.

Degree distribution

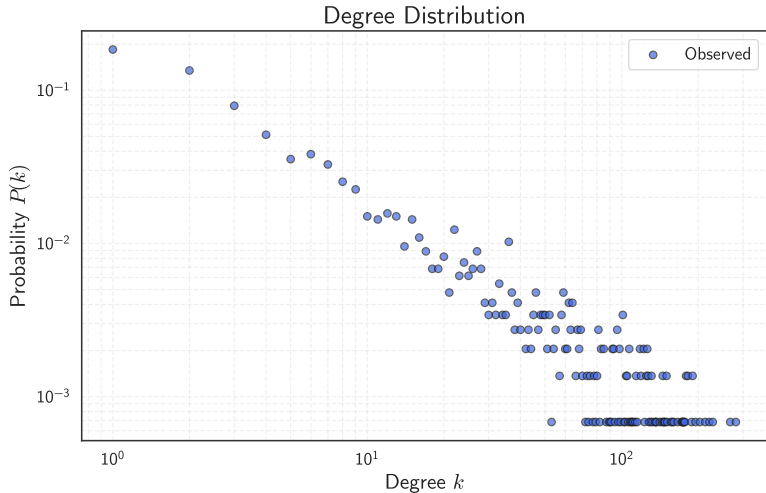


Figure: Degree distribution in linear binning.

Degree distribution

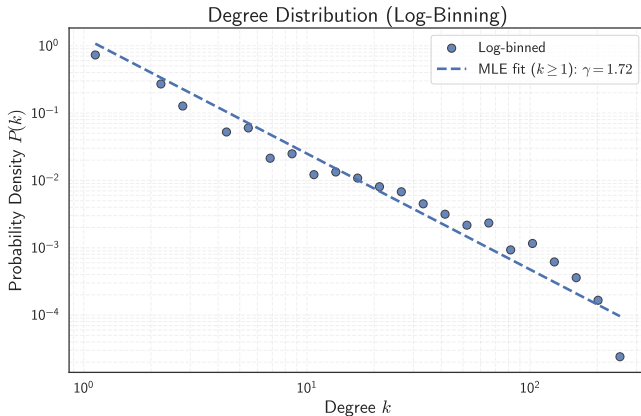


Figure: Degree distribution with logarithmic binning. Using MLE, we find a power-law behavior with $\gamma = 1.722$. This is comparable to the Abe–Suzuki network: $\gamma = 1.33$ for Southern California and $\gamma = 2.22$ for Japan (both computed with a cell size of 10 km).

Network Structural Properties

- **Giant Component**

- Total components: 354
- Largest component: 1,436 nodes (79.9% of the network)
- $\Rightarrow$ Most nodes belong to a single connected structure

- **Clustering Coefficient**

- Average clustering: $C = 0.391$
- Erdős–Rényi baseline: $C_{ER} \approx \langle k \rangle / N = 0.015$
- $\Rightarrow$ Strong local clustering compared to random networks

- **Path Length and Diameter**

- Average path length: $L = 3.25$
- Random expectation: $L_{ER} \approx \frac{\ln N}{\ln \langle k \rangle} = 2.38$
- Diameter: 9 $\Rightarrow$ Small-world behavior

- **Adjacency Matrix**

- Sparsity: 99.36% $\Rightarrow$ The network is highly sparse

Centrality

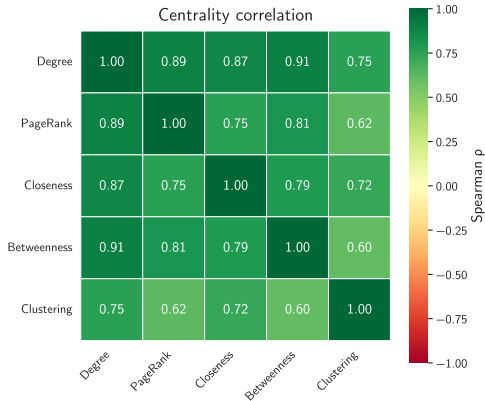


Figure: Centrality Spearman ρ correlation matrix between different centrality metrics.

Degree, PageRank and betweenness agree ($\rho \approx 0.85$ - 0.91): the same cells are central under every measure. Clustering is the outlier.

Top centrality agreement

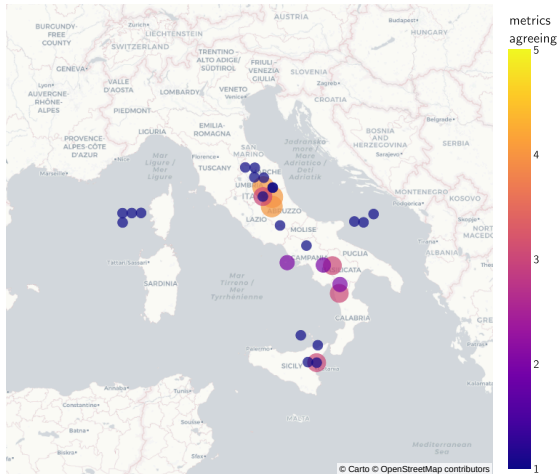
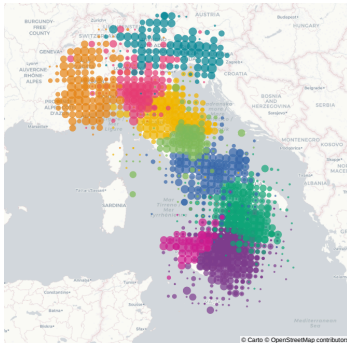


Figure: Cells ranked in the top 10 across the five centrality measures. Four metrics agree only on the central Apennines (Abruzzo); the southern Apennines (Campania–Basilicata) and Etna form weaker clusters.

Community detection 1

Seismic Communities: Louvain



Seismic Communities: Louvain (consensus)

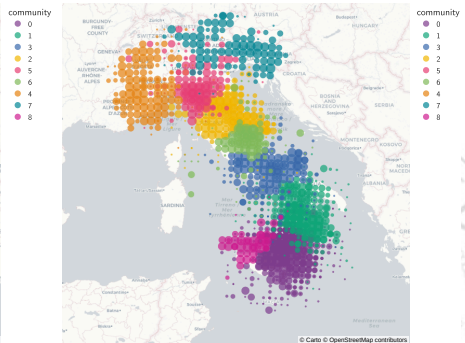
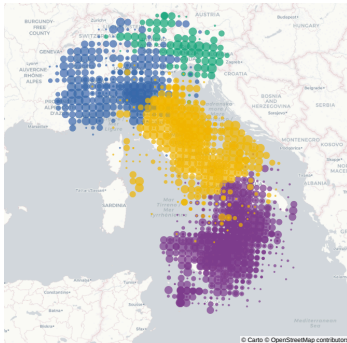


Figure: Louvain and consensus Louvain give identical results ($NMI = 1.0$): the partition is stable, not an artifact of one random seed. Only communities with more than 10 cells shown.

Community detection 2

Seismic Communities: InfoMap



Seismic Communities: HDBSCAN (geo, density)

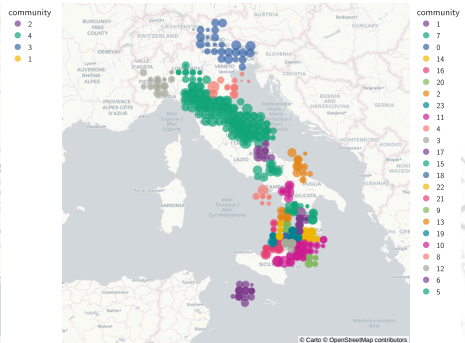


Figure: InfoMap, HDBSCAN-geo.
Only communities with more than 10 cells are shown; this drops HDBSCAN's singleton/noise clusters.

Community metrics

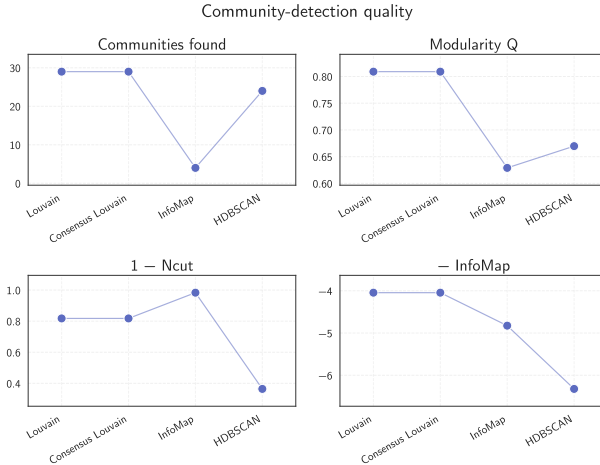


Figure: Number of communities per method, plus three of the four quality measures used (Modularity, Ncut, InfoMap; NMI on the next slide). All axes oriented so higher = better. No method wins on all.

Community NMI

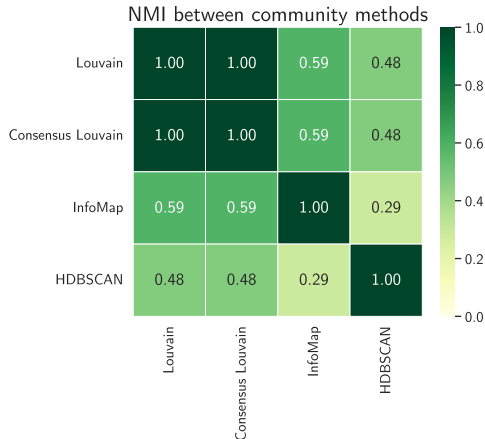


Figure: NMI between community methods. Louvain and consensus Louvain are identical. Agreement with the geography-only HDBSCAN baseline is low / moderate (0.29–0.48): the topology methods capture seismic flow, not just location.

Communities vs DISS sources

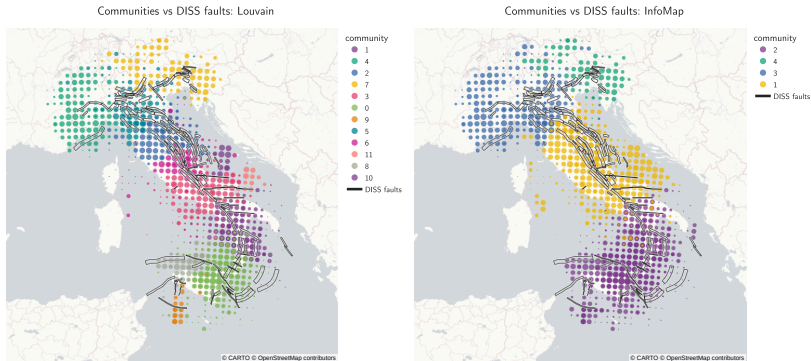
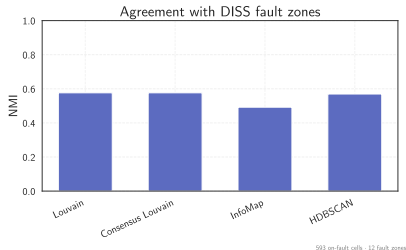


Figure: Detected communities (Louvain, InfoMap) overlaid on the DISS sources. Community boundaries track the mapped Apennine fault systems: the network recovers tectonic segmentation.

Community - fault agreement



- DISS fault zones give us an **independent check** (12 zones, 593 on-fault cells).
- $\Rightarrow$ **link-only** methods (Louvain, InfoMap) reach ≈ 0.5 – 0.6 : they match **real faults** without ever seeing the map.

Known sequences - Louvain

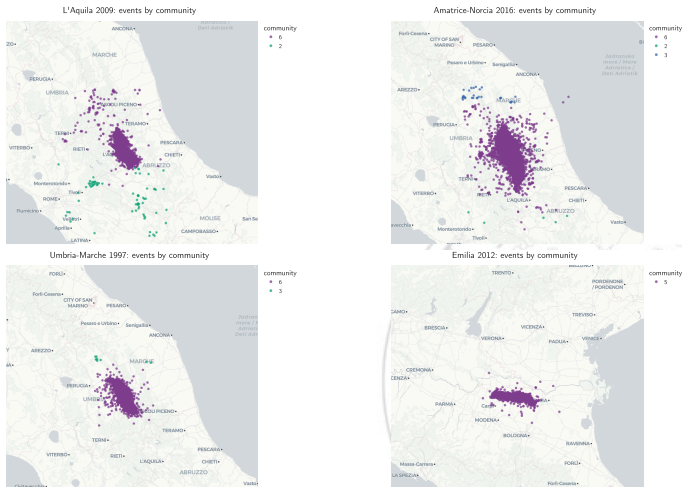


Figure: Louvain recovers each sequence (L'Aquila 2009, Amatrice–Norcia 2016, Umbria–Marche 1997, Emilia 2012) as predominantly one community: detection isolates real aftershock sequences.

Known sequences - InfoMap

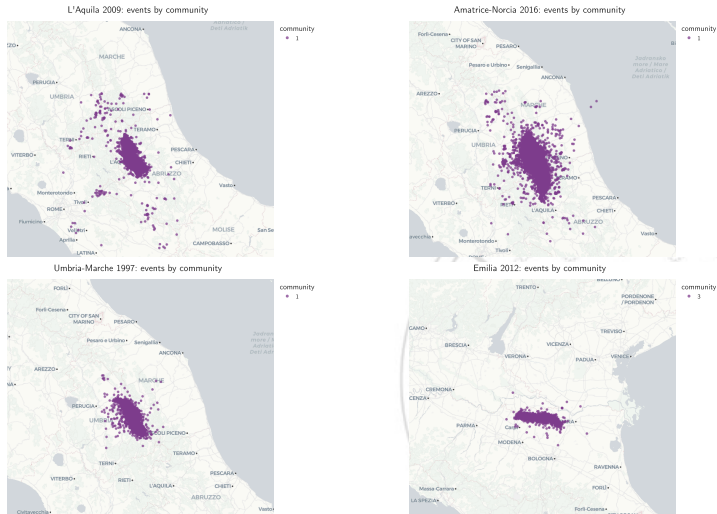


Figure: InfoMap also isolates each sequence as a single community, confirming the recovery seen with Louvain.

Known sequences - HDBSCAN

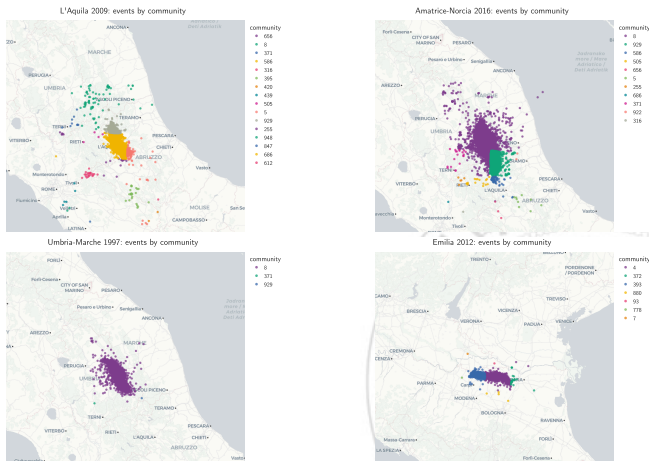


Figure: HDBSCAN isolates only Umbria–Marche 1997 as a single community, while the other three sequences are split into several. This explains the earlier fault-NMI score: HDBSCAN wins on spatial compactness, not by recovering sequences as coherent units.

Modularity Evolution: Amatrice

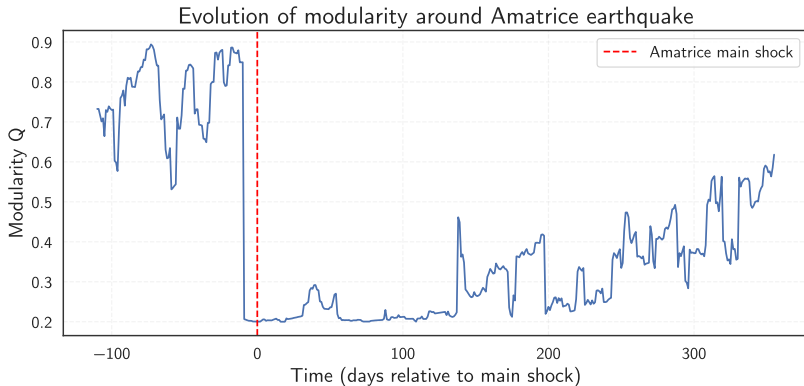


Figure: Time evolution of modularity $Q(t)$ around the Amatrice main shock.

Modularity Evolution: Amatrice

The results are in line with what Abe and Suzuki found.

- **Before the main shock:**
 - High modularity Q , well-defined communities
 - Seismic activity localized in weakly interacting regions
- **During the main shock:**
 - Sharp drop in Q and collapse of community structure
 - Global reorganization, merging of communities
- **After the main shock:**
 - Gradual recovery of Q
- **Conclusion:**
 - Main shocks act as global reconfiguration events



Degree Assortativity

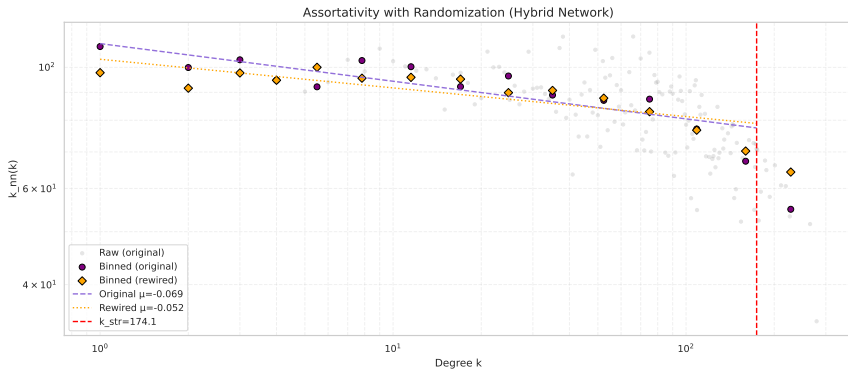
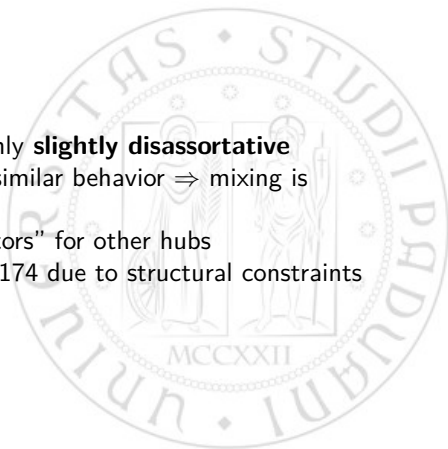


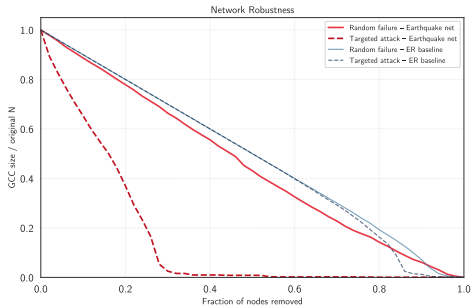
Figure: $k_{nn}(k)$ vs k (log-log), with structural cutoff k_{str} highlighted. In orange, we performed a degree-preserving randomization.

Assortativity Results

- **Results (intrinsic regime $k < k_{\text{str}}$):**
 - $\mu_{\text{orig}} = -0.069$
 - $\mu_{\text{rand}} = -0.053$
 - $|\Delta\mu| = 0.017$
- **Interpretation:**
 - Network is overall neutral, only **slightly disassortative**
 - Randomized network shows similar behavior $\Rightarrow$ mixing is largely **structure-driven**
 - Hubs are not “special attractors” for other hubs
 - Small deviations near $k_{\text{str}} \approx 174$ due to structural constraints



Robustness



- **Random failure:** slow, near-linear decay (like ER) $\Rightarrow$ resilient
- **Targeted attack:** removing $\sim 30\%$ of hubs shatters the network
- **Robust-yet-fragile** scale-free signature.
- Connectivity relies on a few hub cells along the most active fault segments.

Conclusions

- **Topology:** scale-free network ($\gamma \approx 1.722$), small-world ($C = 0.391 \gg C_{ER} = 0.015$, $L = 3.25$ comparable with $L_{ER} = 2.38$)
- **Hubs:** found on DISS faults; centralities agree ($\rho \approx 0.85\text{--}0.91$)
- **Communities:** seismotectonic provinces, beyond geography (geo-baseline NMI ≈ 0.49); consistent with DISS
- **Dynamics:** main shock $\Rightarrow$ modularity collapse ($Q : 0.85 \rightarrow 0.2$), gradual recovery

References

1. S. Abe and N. Suzuki, *Scale-free network of earthquakes*, Europhysics Letters (2003).
2. S. Abe and N. Suzuki, *Complex-network description of seismicity*, Nonlinear Processes in Geophysics (2006).
3. S. Abe and N. Suzuki, *Dynamical evolution of the community structure of complex earthquake networks*, Europhysics Letters (2012).
4. D. Leon, J. Valdivia, V. Bucheli, *A revision of seismicity models based on complex systems and earthquake networks*, Journal of Seismology (2021).
5. Earthquakes Dataset taken from *Istituto Nazionale di Geofisica e Vulcanologia* (INGV).
6. DISS Working Group (2025), *Database of Individual Seismogenic Sources (DISS)*, Version 3.3.1, INGV.

Thanks for your attention

Complements

Spearman correlation

- Pearson correlation of the **ranks**:

$$\rho = 1 - \frac{6 \sum_i d_i^2}{n(n^2 - 1)}, \quad d_i = \text{rank}(x_i) - \text{rank}(y_i).$$

- Detects any **monotonic** relation; robust to heavy tails
- $\rho \in [-1, 1]$ (rank agreement)

